

● HARD

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

02

10 questions · ~15 min

Q1

GRID-IN

GEOMETRY AND TRIGONOMETRY

Points A and B lie on a circle with radius 1, and arc $\overline{AB}$ has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc $\overline{AB}$?

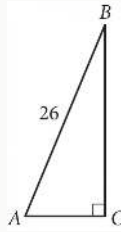
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN GEOMETRY AND TRIGONOMETRY

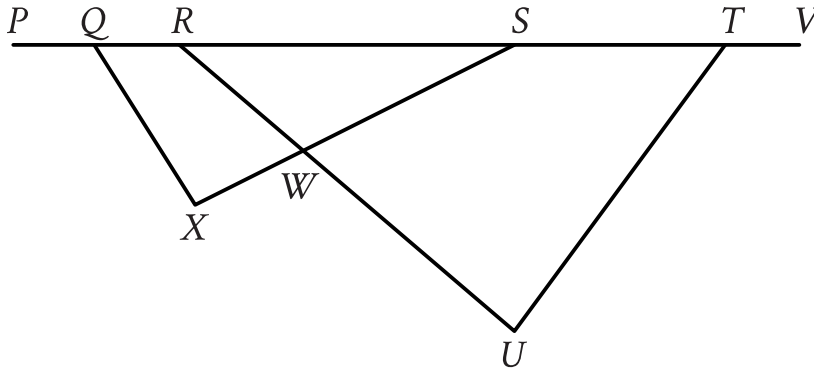


Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

STUDENT-PRODUCED RESPONSE

☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐
☐	☐	☐	☐

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- From left to right, horizontal line segment upper P upper V includes the following points:
 - Upper P
 - Upper Q
 - Upper R
 - Upper S
 - Upper T
 - Upper V
- Points upper X and upper U are each below line segment upper P upper V.
- From left to right, the following 2 overlapping triangles are formed between points on line segment upper P upper V and points upper X and upper U:
 - Triangle upper Q upper X upper S
 - Triangle upper R upper U upper T
- Line segments upper R upper U and upper S upper X intersect at point upper W.
- A note indicates the figure is not drawn to scale.

In the figure shown, points Q , R , S , and T lie on line segment PV , and line segment RU intersects line segment SX at point W . The measure of $\angle SQX$ is 48° , the measure of

$\angle SXQ$ is 86° , the measure of $\angle SWU$ is 85° , and the measure of $\angle VTU$ is 162° . What is the measure, in degrees, of $\angle TUR$?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Rectangle ABCD is similar to rectangle EFGH. The area of rectangle ABCD is 648 square inches, and the area of rectangle EFGH is 72 square inches. The length of the longest side of rectangle ABCD is 36 inches. What is the length, in inches, of the longest side of rectangle EFGH?

- A. 4
- B. 9
- C. 12
- D. 36

Q5

GRID-IN

GEOMETRY AND TRIGONOMETRY

The equation $(x+6)^2 + (y+3)^2 = 121$ defines a circle in the xy -plane. What is the radius of the circle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Triangle ABC is similar to triangle DEF , where A corresponds to D and C corresponds to F . Angles C and F are right angles. If $\tan A = \sqrt{3}$ and $DF = 125$, what is the length of DE ?

A. $125\frac{\sqrt{3}}{3}$

B. $125\frac{\sqrt{3}}{2}$

C. $125\sqrt{3}$

D. 250

GRID-IN GEOMETRY AND TRIGONOMETRY

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

GEOMETRY AND TRIGONOMETRY — SET 2 · GEOMETRY AND TRIGONOMETRY

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 58 square centimeters cm^2 , and the surface area of right rectangular prism Y is 1,450 cm^2 . The volume of right rectangular prism Y is 1,250 cubic centimeters cm^3 . What is the sum of the volumes, in cm^3 , of right rectangular prism X and right rectangular prism Y?

STUDENT-PRODUCED RESPONSE

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.	/	.	/	.	.
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x + 4^2 + y - 19^2 = 121$$

The graph of the given equation is a circle in the xy -plane. The point a, b lies on the circle. Which of the following is a possible value for a ?

- A. -16
- B. -14
- C. 11
- D. 19

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY

A triangle with angle measures 30° , 60° , and 90° has a perimeter of $18 + 6\sqrt{3}$. What is the length of the longest side of the triangle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.